previous chapters. The chapter also rounds out your metaprogramming training with two new topics: the somewhat controversial eval method and the callback methods that you can use to intercept events in the object model.

Part II of the book, *Metaprogramming in Rails*, is a case study in metaprogramming. It contains short chapters that focus on different areas of Rails, the flagship Ruby framework. By looking at Rails' source code, you'll see how master Ruby coders use metaprogramming in the real world to develop great software, and you'll also understand how some metaprogramming techniques evolved in the last few years.

Three appendixes close the book. Appendix 1, *Common Idioms*, on page 217, is a grab-bag of common techniques that are not explained anywhere else in the book. Appendix 2, *Domain-Specific Languages*, on page 227, is a quick look at a programming approach that is common among Ruby developers. Appendix 3, *Spell Book*, on page 231, is a catalog of all the spells in the book, complete with code examples.

"Wait a minute," I can hear you saying. "What the heck are *spells*?" Oh, right, sorry. Let me explain.

Spells

This book contains a number of metaprogramming techniques that you can use in your own code. Some people might call these *patterns* or maybe *idioms*. Neither of these terms is very popular among Rubyists, so I'll call them *spells* instead. Even if there's nothing magical about them, they *do* look like magic spells to Ruby newcomers.

You'll find references to spells everywhere in the book. I reference a spell with the convention *Class Macro (117)* or *String of Code (141)*, for example. The number in parentheses is the page where the spell receives a name. If you need a quick reference to a spell, you'll find it in Appendix 3, *Spell Book*, on page 231.

Quizzes

Every now and then, this book also throws a quiz at you. You can skip these quizzes and just read the solution, but you'll probably want to solve them on your own just because they're fun.

Some quizzes are traditional coding exercises; others require you to get off your keyboard and think. All include a solution, but most quizzes have more than one possible answer. Please, feel free to go wild and experiment.